

Unit 13, Lesson 1: Parallel and Perpendicular Lines



Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to classify parallel and perpendicular lines.



13.1.1 Warm-Up: Artificial Intelligence and Machine Learning

Artificial intelligence enables a machine to simulate human behavior and decisions. Machine learning is when a machine uses artificial intelligence to learn from past data without a programmer having to re-program the machine.

1. Describe one task you would like a machine to be programmed to do.

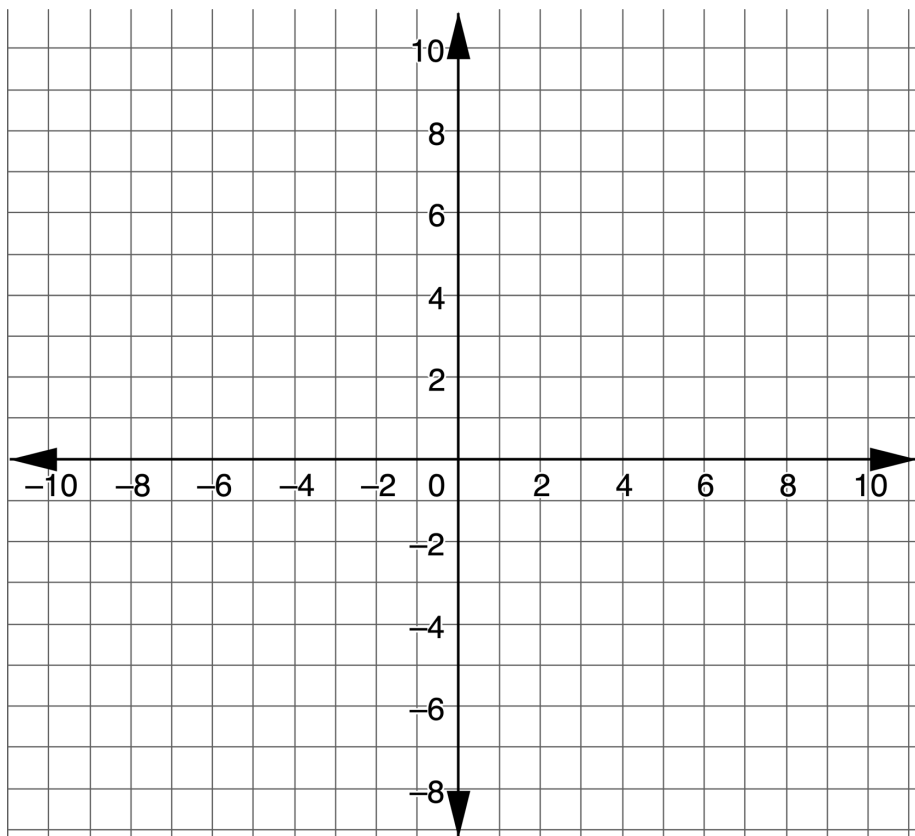
Both artificial intelligence and machine learning rely on mistakes to help improve the programming.

2. Describe a mistake you have made that helped you improve.



13.1.2 Collaboration: Parallel and Perpendicular Lines

1. Graph the equation $y = -2x + 5$ on the coordinate plane.



2. Rewrite the equation $6x + 3y = -3$ in slope-intercept form.

$y =$ _____

3. Graph the equation $6x + 3y = -3$ using slope-intercept form.

4. What similarities do the two equations have?

5. What differences do the two equations have?

6. Complete the statement.

The equations $y = -2x + 5$ and $6x + 3y = -3$ resulted in

☐ parallel
☐ perpendicular
☐ vertical

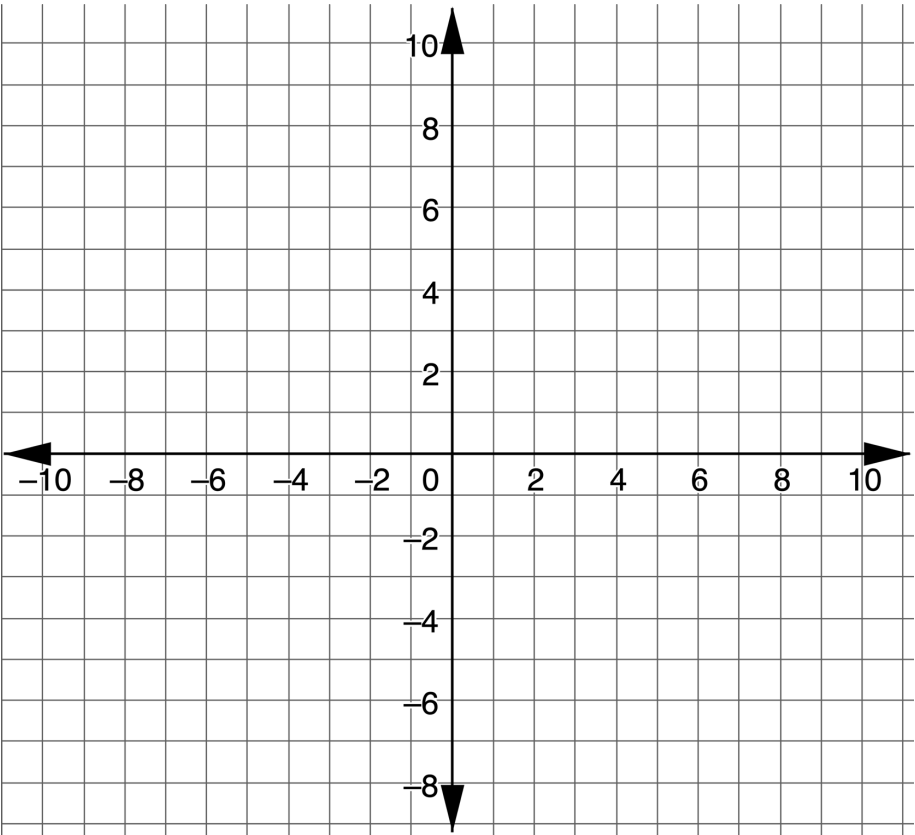
lines. The slopes of the equations are

☐ equal.
☐ undefined.
☐ opposite reciprocals.

7. Rewrite the equation $4y - 6x = 8$ in slope-intercept form.

$y =$ _____

8. Graph the equation $4y - 6x = 8$ using slope-intercept form.



9. Graph the equation $y = -\frac{2}{3}x - 4$.
10. What do you notice about the slopes in the two equations?

11. Complete the statement.

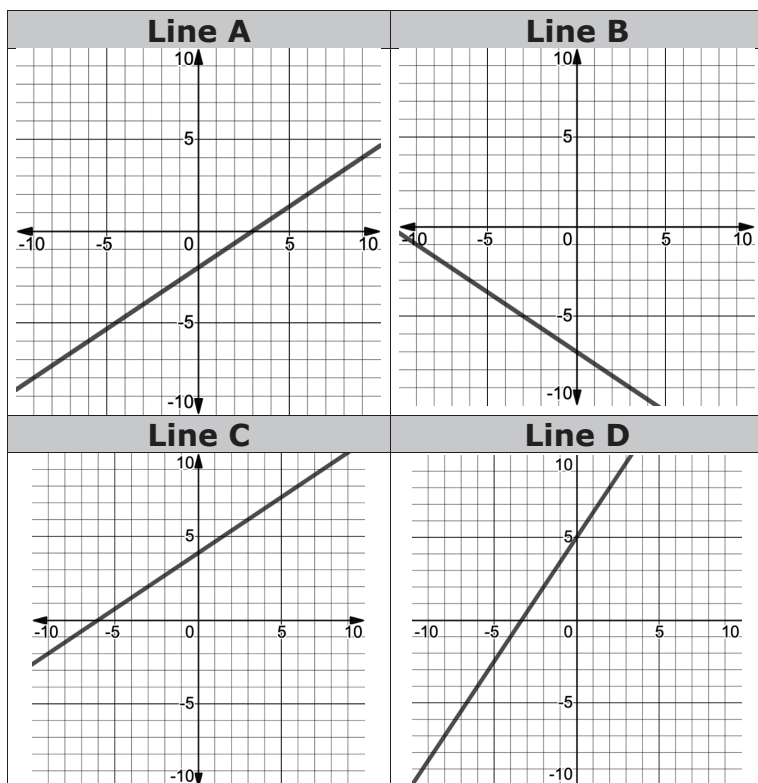
Equation $4y - 6x = 8$ and equation $y = -\frac{2}{3}x - 4$ form
☐ parallel
☐ perpendicular
☐ vertical lines; therefore, the slopes of the
equations are ☐ equal.
☐ undefined.
☐ opposite reciprocals.

12. Complete the table.

Type	Sketch	Slope
Parallel lines		
Perpendicular lines		

13. Match the two lines that are parallel and the two lines that are perpendicular. Then, match the equations.

	Lines	Equations
Parallel		
Perpendicular		



Equation E	Equation F
$y + 2 = \frac{2}{3}(x + 9)$	$2x + 3y = -21$
Equation G	Equation H
$-2x + 3y = -6$	$y = \frac{3}{2}x + 5$



13.1.3 Guided Instruction: Parallel and Perpendicular Lines

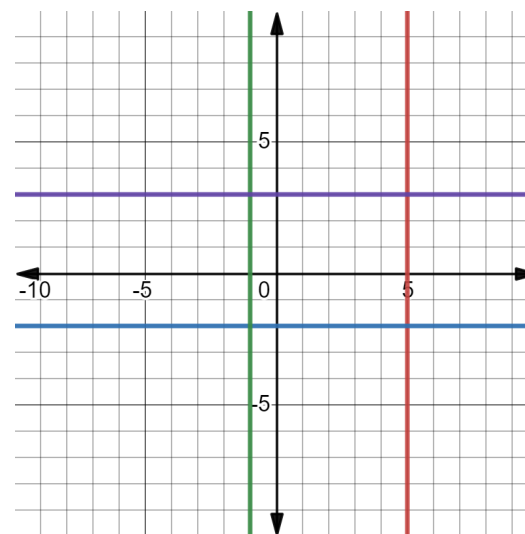


Lines are **parallel** if the slopes of the lines are the same.



Lines are **perpendicular** if the slopes are opposites and reciprocals.

1. Four lines are shown on the coordinate plane.

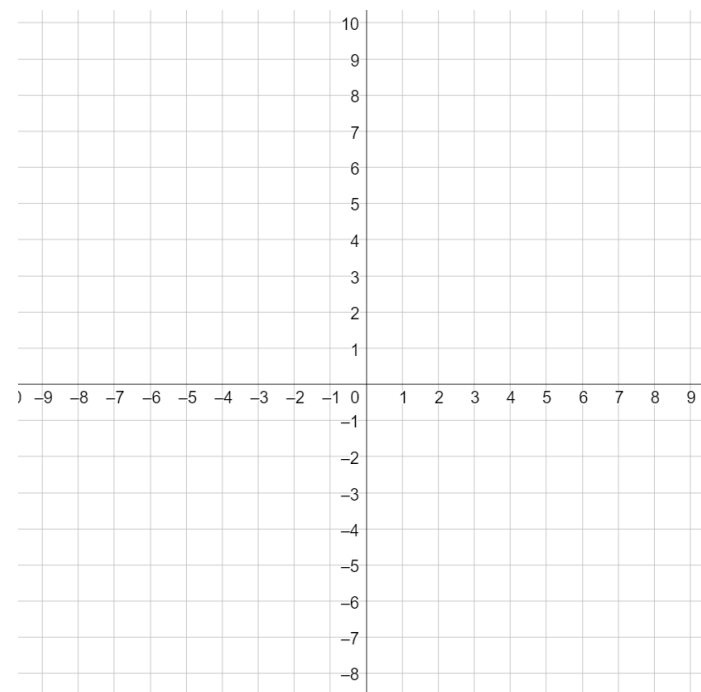


- a. What do you notice about the four lines?

- b. In the graph of the four lines, the four lines create a rectangle. Explain how each of the properties of a rectangle could be shown.

Property	Explanation
Opposite sides are parallel.	
All angles are right angles.	

2. Line W has points $(-1,6)$ and $(1,2)$ on it. Line T has points $(-4,3)$ and $(-2,-2)$ on it.



- a. Graph line W and line T .
- b. Determine if line W and line T are parallel lines, perpendicular lines, or neither. Justify your answer with math.

3. Line N is written as $y = \frac{1}{3}x - 8$. Line V is written as $6x + 2y = 4$.

Determine if line N and line V are parallel lines, perpendicular lines, or neither. Justify your answer with math.



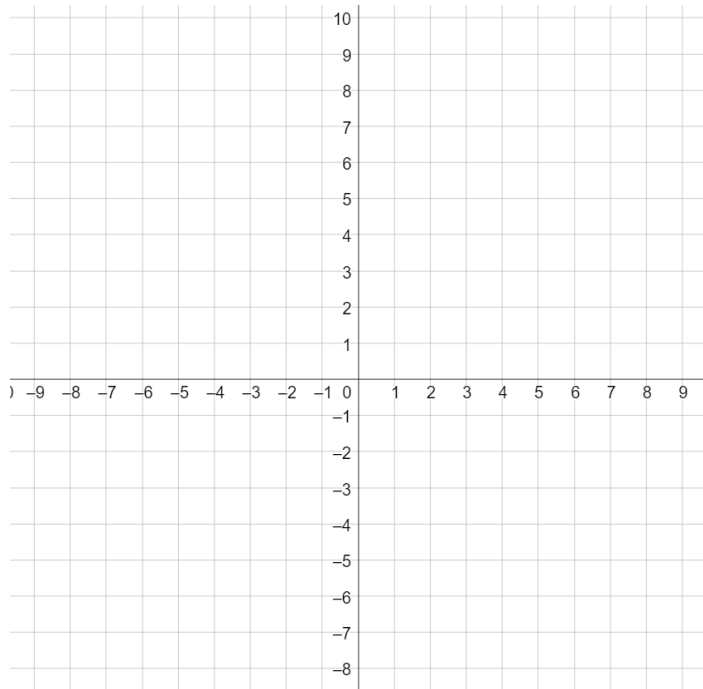
13.1.4 Guided Practice: Parallel and Perpendicular Lines

1. Given that line M is $3x - y = 24$, determine if the following lines are parallel to line M .

Complete the table.

Equation of Line	Work	Is the line parallel to line M ?
$y = 3x + 8$		☐ Yes ☐ No
$y = -3x - 1$		☐ Yes ☐ No
$3y = x - 4$		☐ Yes ☐ No
$3x - y = 3$		☐ Yes ☐ No

2. Line R is written as $4x + y = 5$, line K is written as $12y = 4x - 12$, and line C is written as $4y - 12 = x$.



- Graph lines R , K , and C .
- Determine which two lines are perpendicular lines. Justify your answer with math.



13.1.5 Wrap-Up: Cool Down

- Write a short note to a friend who missed today's class summarizing what you learned.